# General Mathematics — Exercise Sheet 28

*Integrals in geometry and physics*

**Variant: Statements**

**Exercise 1 (Polar area and length).** Fix $a > 0$. Consider the polar curve with radius $r(\varphi) = a(1 + \cos\varphi)$ for $0 \le \varphi \le 2\pi$.

(a) Use Lecture 28's polar-sector formula to compute the enclosed area. Explain why this angular interval counts the interior once.

(b) Starting from $x = r\cos\varphi$ and $y = r\sin\varphi$, derive the polar length formula $L = \int_\alpha^\beta \sqrt{r^2 + (r')^2}\,\mathrm{d}\varphi$ for a $C^1$ radius on $[\alpha, \beta]$. Use the component derivatives of Lecture 8 and the length theorem of Lecture 28.

(c) Compute the full curve length. Retain the absolute value when taking $\sqrt{4a^2\cos^2(\varphi/2)}$. Explain why the zero derivative at $\varphi = \pi$ does not invalidate the theorem.

**Exercise 2 (A ball with an axial hole).** A ball of radius 2, centred at the origin in $\mathbb{R}^3$, has a cylindrical hole of radius 1 drilled along the $x$ axis. The remaining solid is

$$D = \{(x, y, z) : x^2 + y^2 + z^2 \le 4,\ y^2 + z^2 \ge 1\}.$$

Determine the interval of $x$ values with nonempty cross-sections and compute the remaining volume. Use the disc and washer formulas of Lecture 28.

**Exercise 3 (An unbounded rod density).** Let $L > 0$, $\kappa > 0$ and $p \in \mathbb{R}$. On $0 < x \le L$, let the linear mass density be $\lambda(x) = \kappa x^p$. Interpret every integral at zero by Lecture 27's separate finite-cutoff limit.

(a) Determine separately when the total mass $m$, the first moment $Q_0$ and the endpoint inertia $I_0$ are finite, and compute each finite quantity.

(b) Whenever $m$ is finite, compute the centre coordinate $\bar{x}$ and central inertia $I_{\bar{x}}$. Justify the parallel-axis identity by finite-cutoff expansion and convergent limits.

The definitions of mass and moments are those of Lecture 28. The constant $\kappa$ has the units needed to make $\lambda$ mass per unit length.

**Exercise 4 (Mass and axial inertia of a disc).** A thin circular disc of radius $a > 0$ has surface mass density $\sigma(r) = \sigma_0(1+r^2/a^2)$, where $r$ is distance from the centre and $\sigma_0 > 0$ is mass per unit area. Derive annular-piece integrals for its mass and its moment of inertia about the perpendicular axis through the centre, and evaluate them. Use polar area from Lecture 28 and the squared-distance definition of inertia.

**Exercise 5 (Surface area at a singular graph endpoint).** Use the following surface-of-revolution formula as a given [C] here: a nonnegative $C^1$ graph $y = f(x)$ on $[a, b]$, with $a < b$, rotated about the $x$ axis, has lateral area $S = 2\pi \int_a^b f(x)\sqrt{1 + f'(x)^2}\,\mathrm{d}x$. For an excluded endpoint, define its lateral area by the limit of these finite-cutoff areas when that limit is finite. Compute the lateral area generated by $f(x) = \sqrt{2x}$ for $0 < x \leq 2$. This means the curved surface, without adding an end disc. The given formula is Stewart, §8.2; Lecture 41 develops surface area.

**Exercise 6 (Escape work in an inverse-square model).** Assume a body of mass $m > 0$ moves radially outside a spherical source of mass $M > 0$ and radius $R > 0$. Take as physical givens the gravitational force magnitude $GMm/r^2$, the kinetic energy $mv^2/2$, and conservation of kinetic energy plus work against gravity. The constant $G$ is positive. Neglect resistance and other forces.

(a) Compute the external work needed to lift the body slowly from $r = R$ to arbitrarily large radius.

(b) Determine the least initial outward speed $v_0 \geq 0$ whose energy permits reaching every finite radius without further external work. Check the equality case by computing the remaining kinetic energy at radius $r$.

Use Lecture 28's signed work and Lecture 27's improper-limit definition.

**Exercise 7 (Mass measured along a spatial curve).** A thin wire follows the parametrisation

$$\boldsymbol{r}(t) = (a\cos t, a\sin t, bt), \qquad 0 \leq t \leq 2\pi,$$

where $a > 0$ and $b > 0$. Its linear density per unit arc length at parameter $t$ is $\lambda(t) = \lambda_0(1 + t/(2\pi))$, with $\lambda_0 > 0$. Find the wire length and its total mass. Explain why the mass integrand contains the arc-length factor rather than just $\lambda(t)$.